ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

ELECTRICAL TECHNOLOGY

THEORY PAPER

MARKS: 100 **TIME: 2 hours**

Topics: safety, tools and measuring instruments, DC circuits, basic electronics and digital logic.
Suitable for Electrical, Electronics and Digital Electronics learners.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show formulae, substitutions and units in ALL calculations.

QUESTION 1: MULTIPLE CHOICE

1.1 The unit of resistance is the …

A volt. B ampere.
C ohm. D watt.

1.2 The instrument used to measure potential difference is the …

A ammeter. B voltmeter.
C ohmmeter. D wattmeter.

1.3 A resistor has the colour bands brown, black, red, gold. Its value is …

A 100 Ω. B 1 kΩ.
C 10 kΩ. D 102 Ω.

1.4 A diode allows current to flow …

A in both directions. B in one direction only.
C only when it is AC. D never.

1.5 The logic gate whose output is 1 only when ALL its inputs are 1 is the …

A OR gate. B AND gate.
C NOT gate. D NOR gate.

(5 × 2)
[10]

QUESTION 2: SAFETY

2.1 Give THREE causes of electric shock. (3)

2.2 Describe what you should do if you find a person being electrocuted. (3)

2.3 What is the purpose of an earth leakage unit in a distribution board? (2)

2.4 Give FOUR safety rules for an electrical workshop. (4)

[12]

QUESTION 3: TOOLS AND INSTRUMENTS

3.1 State the use of: side cutters, long-nose pliers, wire strippers, soldering iron. (4)

3.2 Explain how a multimeter must be connected to measure current, and give ONE precaution. (3)

3.3 Give THREE features of a good soldered joint. (3)

3.4 Determine the value and tolerance of a resistor with the colour bands yellow, violet, orange, gold. (3)

[13]

QUESTION 4: DC CIRCUITS

4.1 State Ohm's law. (2)

4.2 Three resistors of 10 Ω, 20 Ω and 30 Ω are connected in series to a 12 V supply.

4.2.1 Calculate the total resistance. (2)

4.2.2 Calculate the current in the circuit. (3)

4.2.3 Calculate the potential difference across the 20 Ω resistor. (3)

4.2.4 Calculate the total power used by the circuit. (3)

4.3 Calculate the total resistance of two 10 Ω resistors connected in parallel. (2)

4.4 State Kirchhoff's current law. (2)

4.5 Two 1,5 V cells are connected (a) in series and (b) in parallel. Write down the total voltage in each case, and give ONE advantage of connecting cells in parallel. (4)

4.6 Explain TWO differences between alternating current (AC) and direct current (DC). (4)

[25]

QUESTION 5: ELECTRONICS

5.1 Explain what is meant by forward bias of a diode, and what happens to the current. (4)

5.2 Why must an LED always be connected with a resistor in series? (2)

5.3 An LED needs 2 V and 20 mA. It is connected to a 9 V battery. Calculate the value of the series resistor needed. (3)

5.4 What is the main function of a capacitor? (2)

5.5 Explain how the resistance of a light-dependent resistor (LDR) changes with light, and give ONE use. (3)

5.6 Name TWO types of switches. (2)

5.7 Give ONE difference between a fuse and a circuit breaker. (2)

5.8 How does the resistance of an NTC thermistor change when its temperature increases? (2)

[20]

QUESTION 6: DIGITAL LOGIC

6.1 Convert 13 (decimal) to binary. (2)

6.2 Convert 10110_2 to decimal. (2)

6.3 Draw the truth tables for a two-input AND gate and a two-input OR gate. (6)

6.4 Describe the function of a NOT gate. (2)

6.5 Write down the Boolean expressions for a two-input AND gate and a two-input OR gate (inputs A and B, output Y). (2)

6.6 The output of an AND gate is fed into a NOT gate. Name the combined gate and draw its truth table. (4)

6.7 Give TWO reasons why digital systems use the binary number system. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER